2: Air resistance and the drag coefficient C_d

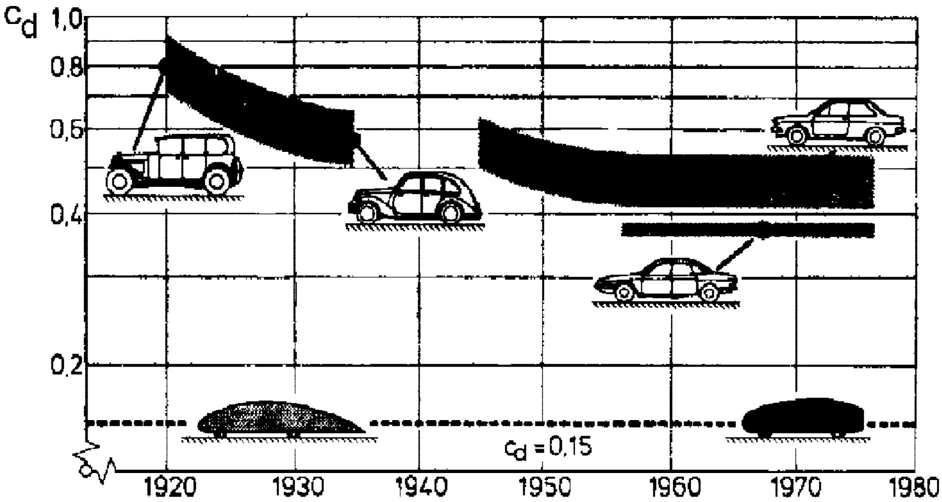

Figure 2.1: The value of C_d only improved rather slowly until the energy crisis in the 1970-es. The smallest values illustrated are considered as a lower limit for a car, given that it should be reasonably comfortable in practical use.

After the energy crisis in the 1970-es, the so-called drag coefficient C_d for a car became an important parameter in the motor magazines test reports. A more streamlined shape gives a smaller value of C_d and less fuel is needed to overcome the air resistance. Fig. 2.1 illustrates how the value of C_d has improved from about 0.85 in the early cars to just under 0.40 in the early 1980-es. Today a well shaped family car has $C_d \simeq 0.30$ and most big motor companies do experiments on *hyper cars* that come close to the lower limit of $C_d \simeq 0.15$ mentioned in the text to fig. 2.1. A simple search on *Google* under *Hyper cars* gives many more examples than those shown here. Still smaller values are reached with the solar powered cars participating in competitions. See fig. 2.3.

Figure 2.2: General Motors EV1. This so-called *hyper car* is an experimental car with a value of C_d as low as 0.19.

Figure 2.3: The Volkswagen L1 (or "1 Litre car") that, with $C_d = 0.16$, goes 100 kilometres per litre

Figure 2.4: *Oct. 22, 2003: The Dutch solar car Nuna II, using ESA space technology, finished first in the World Solar Challenge, a 3010 km race right across Australia for cars powered by solar energy.* [1,2,3,4]. It covered this distance with a new record for the average speed: 97 km/h. The car was on show at *Physics on Stage* 2003 where the present author was told that its drag coefficient was as low $C_d = 0.08$.

The dimenssionless drag coefficient is defined by

$$C_d = \frac{F}{\frac{1}{2}\varrho A v^2} \tag{2.1}$$

or, equivalently,

$$F = \frac{1}{2}\rho A v^2 \cdot C_d \tag{2.2}$$

where v is the velocity, ϱ is the density of the air and A is the cross sectional area, also called the *frontal area*. This latter is defined a shown in fig. 2.5, and it can be determined as explained in appendix E. The density of the air is given by

$$\varrho = 1.29 \frac{kg}{m^3} \cdot \frac{p}{p_0} \cdot \frac{T_0}{T} \tag{2.3}$$

where $p_0 = 1000\,hPa$ is the normal value of the atmospheric pressure p and $T_0 = 273\,K$ is the Kelvin temperature T at $0\,^oC$.

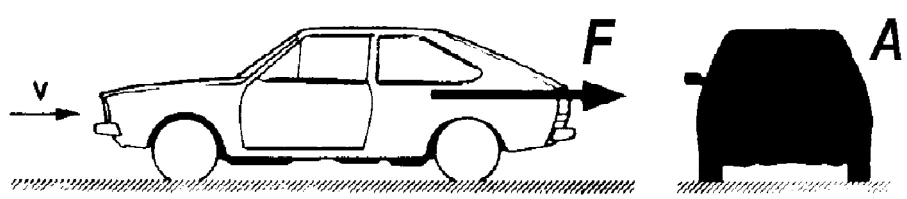

Figure 2.5: The car is placed in a wind tunnel where air is passed through with the velocity v. The drag force F is proportional to the frontal area A, illustrated to the right.

The study of air resistance is course of a great interest not only for cars. The drag coefficient is defined as in (2.1) for any that body moves through the air, while keeping a constant geometric form as well as a constant orientation relative to the direction of the velocity. The standard method for determining C_d for a given body at a given velocity v is to place it in a wind tunnel and measure the drag caused by an air stream of that velocity.

By the definition in (2.1), it is not surprising that the graph for C_d as a function of v always diverges toward infinity in the limit $v \to 0$. It is, however, an empirical fact, that as v increases, the graph soon settles so that it only varies a little about a certain constant value of the order of magnitude[1] $C_d \simeq 1$. Here it stays up to a rather high velocity, where, sometimes, it can begin to vary again. The C_d -values for the cars considered above are only constants which approximate the exact velocity-dependent values of $C_d(v)$ over the speeds v by which cars normally travel.

An important part of the present notes are about the possibility of finding C_d for a given body by measuring the air resistance on a similar model. What is the connection between C_d for the full scale body and C_d for a smaller model? As we shall see in sec. 8, the similarity means that

the two graphs settle close to the same constant C_d -value. (2.4)

This is the background for the results collected in the table in appendix F. Here a constant C_D is assigned to various geometric forms - a sphere, for instance, is assigned $C_D = 0,47$, whatever its size. A circular cylinder, moving horizontally with the axis vertical, is assigned $C_d = 0.63$ if height = diameter, $C_d = 0.83$ if height = $2 \times$ diameter, etc., independently of the absolute size of the cylinder. Had these simple results been true without reservations, C_d for a given object could always be determined by placing a small easily handled model in a wind tunnel. There *are*, however, reservations: The velocity intervals, within which $C_d \simeq$ constant, are not independent of size. In table F1 these intervals are given

[1] This order of magnitude can be understood as follows. If the frontal area A had moved as piston with velocity v, this piston would have covered a volume $A v \Delta t$ in a brief interval of time Δt. The air in this volume would have had the mass $\varrho A v \Delta t$ and if the piston had transferred its own speed v to each little part of this mass, it would have delivered the kinetic energy $\frac{1}{2}(\varrho A v \Delta t) v^2$. Identifying this with the work $F \cdot v \Delta t$ done by the piston, we get the result $F = \frac{1}{2} \varrho A v^2 \cdot 1$.

The low C_d - value of the solar car in fig. 2.4 can then be understood as follows. Only the air particles which pass the cockpit and the guards around the wheels are left with appreciable whirling velocities. The main body just pushes them a little up-down if they pass above the car, and a little down-up if they pass underneath. If the car is long, the can do so by first picking up a small velocity and then delivering it again. Seen from the car, the just flow along its surface. With such an almost *laminar flow*, there is only a little transfer to kinetic energy, but we still have some air resistance due *friction* between the air and the surface of the car. See the discussion in sec. 8.

indirectly in terms of the so-called Reynolds number R_{Rey}. We shall, however, postpone the discussion of these problems until sec. 8.